

Problem 1.1.1

PROMPT

Given the tensors

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

Solve:

SOLUTION

a) $\mathbf{A}_{ii}$

$$\mathbf{A}_{ii} \rightarrow \sum_i^3 \mathbf{A}_{ii} = A_{11} + A_{22} + A_{33} = \text{trace}(\mathbf{A}) = 8$$

b) $\mathbf{A}_{ij}\mathbf{b}_j$

$$\begin{aligned} c_i = \mathbf{A}_{ij}\mathbf{b}_j &\rightarrow \sum_j^3 \mathbf{A}_{ij}\mathbf{b}_j = A_{i1}b_1 + A_{i2}b_2 + A_{i3}b_3 = \\ &= \begin{pmatrix} A_{11}b_1 + A_{12}b_2 + A_{13}b_3 \\ A_{21}b_1 + A_{22}b_2 + A_{23}b_3 \\ A_{31}b_1 + A_{32}b_2 + A_{33}b_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} \end{aligned}$$

c) $\mathbf{A}_{ji}\mathbf{b}_j$

$$\begin{aligned} c_i = \mathbf{A}_{ji}\mathbf{b}_j &\rightarrow \sum_j^3 \mathbf{A}_{ji}\mathbf{b}_j = A_{1i}b_1 + A_{2i}b_2 + A_{3i}b_3 = \\ &= \begin{pmatrix} A_{11}b_1 + A_{21}b_2 + A_{31}b_3 \\ A_{12}b_1 + A_{22}b_2 + A_{32}b_3 \\ A_{13}b_1 + A_{23}b_2 + A_{33}b_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} \end{aligned}$$

Note: results for (b) and (c) are the same by coincidence

d) $\mathbf{b}_i\mathbf{b}_j$

$$\mathbf{A}_{ij} = \mathbf{b}_i\mathbf{b}_j = \begin{pmatrix} b_1b_1 & b_1b_2 & b_1b_3 \\ b_2b_1 & b_2b_2 & b_2b_3 \\ b_3b_1 & b_3b_2 & b_3b_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 4 \end{pmatrix}$$

e) $\mathbf{A}_{ij}\mathbf{A}_{ji}$

$$\alpha = \sum_i^3 \sum_j^3 \mathbf{A}_{ij}\mathbf{A}_{ji} =$$

$$\begin{aligned}
& A_{11}A_{11} + A_{12}A_{21} + A_{13}A_{31} + \\
& A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} + \\
& A_{31}A_{13} + A_{32}A_{23} + A_{33}A_{33} \\
& = 36
\end{aligned}$$

Problem 1.1.2

PROMPT

Given the expressions in index notation, write the corresponding expressions in abstract tensor notation.

SOLUTION

a) $a_i b_i$

$$\alpha = a_i b_i = \sum_i^3 a_i b_i = \mathbf{a} \cdot \mathbf{b}$$

b) $a_i b_k c_k$

$$c_i = a_i \sum_k^3 b_k c_k \rightarrow \mathbf{c} = \mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c})$$

c) $A_{ik} B_{kj}$

$$C_{ij} = A_{ik} B_{kj} =$$

$$\sum_k^3 A_{ik} B_{kj} \rightarrow$$

$$\begin{pmatrix}
A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} & A_{11}B_{12} + A_{12}B_{22} + A_{13}B_{32} & A_{11}B_{13} + A_{12}B_{23} + A_{13}B_{33} \\
A_{21}B_{11} + A_{22}B_{21} + A_{23}B_{31} & A_{21}B_{12} + A_{22}B_{22} + A_{23}B_{32} & A_{21}B_{13} + A_{22}B_{23} + A_{23}B_{33} \\
A_{31}B_{11} + A_{32}B_{21} + A_{33}B_{31} & A_{31}B_{12} + A_{32}B_{22} + A_{33}B_{32} & A_{31}B_{13} + A_{32}B_{23} + A_{33}B_{33}
\end{pmatrix}$$

$$= \mathbf{A} \cdot \mathbf{B}$$

d) $A_{ki} B_{kl} A_{lj}$

Assuming the answer as D_{ij} , since only i and j are non repeating, assume

$$C_{il} = A_{ki} B_{kl} \rightarrow \mathbf{C} = \mathbf{A}^T \cdot \mathbf{B}$$

Then

$$D_{ij} = C_{il} A_{lj} \rightarrow \mathbf{D} = \mathbf{C} \cdot \mathbf{A}$$

Finally

$$\mathbf{D} = \mathbf{A}^T \cdot \mathbf{B} \cdot \mathbf{A}$$

Another approach is:

Assume

$$E_{kj} = B_{kl}A_{lj} \rightarrow E = B \cdot A$$

Then

$$D_{ij} = A_{ki}E_{kj} \rightarrow D = A^T \cdot E$$

Which, of course, results in the same value!

$$D = A^T \cdot B \cdot A$$

e) $a_i A_{jk} b_k$

Let the answer be a matrix D_{ij} .

Let $c_j = A_{jk} b_k$

$$c_j = A_{j1}b_1 + A_{j2}b_2 + A_{j3}b_3 \rightarrow c = A \cdot b$$

then,

$$D_{ij} = a_i c_j \rightarrow \begin{pmatrix} a_1 c_1 & a_1 c_2 & a_1 c_3 \\ a_2 c_1 & a_2 c_2 & a_2 c_3 \\ a_3 c_1 & a_3 c_2 & a_3 c_3 \end{pmatrix} = a \cdot c^T$$

Thus

$$D = a \cdot (A \cdot b)^T$$

f) $a_i A_{ik} b_k$

From the previous question, we know that

$$c_i = A_{ik} b_k \rightarrow c = A \cdot b$$

And, simply

$$\alpha = a_i c_i = a \cdot c$$

The answer is

$$\alpha = a \cdot A \cdot b$$

Problem 1.2.1

PROMPT

Show that $\delta_{ij}\delta_{ji} = 3$.

SOLUTION

$$\delta_{ij}\delta_{ji} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^T = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 3$$

Problem 1.2.2

PROMPT

Find the expression in index notation for

$$\frac{\partial \mathbf{v}}{\partial \mathbf{v}}$$

where $\mathbf{v}$ is a vector.

SOLUTION

$$\mathbf{A} = \frac{\partial \mathbf{v}}{\partial \mathbf{v}} = \begin{pmatrix} \frac{\partial v_1}{\partial v_1} & \frac{\partial v_1}{\partial v_2} & \frac{\partial v_1}{\partial v_3} \\ \frac{\partial v_2}{\partial v_1} & \frac{\partial v_2}{\partial v_2} & \frac{\partial v_2}{\partial v_3} \\ \frac{\partial v_3}{\partial v_1} & \frac{\partial v_3}{\partial v_2} & \frac{\partial v_3}{\partial v_3} \end{pmatrix} \rightarrow A_{ij} = \frac{\partial v_i}{\partial v_j}$$

It is known that

$$\frac{\partial v_l}{\partial v_m} = \begin{cases} 1 & \text{if } l = m \\ 0 & \text{else} \end{cases}$$

Then it is safe to assume that

$$\frac{\partial v_i}{\partial v_j} = \delta_{ij}$$

Problem 1.2.3

PROMPT

With the aid of the Kronecker delta, find the expression in index notation for the trace of $\mathbf{a}\mathbf{b}^T$ where $\mathbf{a}$ and $\mathbf{b}$ are vectors.

SOLUTION

$$\alpha = \text{trace}(\mathbf{a}\mathbf{b}^T) = a_i b_j \delta_{ij}$$

With the contraction property

$$b_j \delta_{ij} = b_i$$

$$\alpha = a_i \cdot b_i = \mathbf{a} \cdot \mathbf{b}$$

Problem 1.2.4

PROMPT

Simplify the expression

$$\frac{\partial^2}{\partial x_i \partial x_j} (x_i x_j)$$

SOLUTION

$$\begin{aligned} \frac{\partial^2}{\partial x_i \partial x_j} (x_i x_j) &= \frac{\partial}{\partial x_i} \left(\frac{\partial x_i}{\partial x_j} x_j + \frac{\partial x_j}{\partial x_j} x_i \right) = \\ &= \frac{\partial}{\partial x_i} (0 + \delta_{jj} x_i) = 3 \frac{\partial}{\partial x_i} (x_i) = 3 \delta_{ii} = 9 \end{aligned}$$

Problem 1.3.1

PROMPT

Show that

$$\epsilon_{ijk} a_i a_j = 0$$

SOLUTION

By brute force...

Let $A_{ij} = a_i a_j$

$$\left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) \cdot \begin{pmatrix} a_1 a_1 & a_1 a_2 & a_1 a_3 \\ a_2 a_1 & a_2 a_2 & a_2 a_3 \\ a_3 a_1 & a_3 a_2 & a_3 a_3 \end{pmatrix} = \begin{pmatrix} +a_2 a_3 - a_3 a_2 \\ -a_1 a_3 + a_3 a_1 \\ +a_1 a_2 - a_2 a_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Note: To represent a third order tensor as an array of matrices may not be standard. This was done mainly for visualizations.

By triviality...

Multiplying both sides by the cartesian basis vector:

$$\epsilon_{ijk} a_i a_j \cdot \hat{e}_i = 0 \cdot \hat{e}_i = 0$$

The expression above is the cross product of the vector $\mathbf{a}$ by itself:

$$\epsilon_{ijk} a_i a_j \cdot \hat{e}_i \rightarrow \mathbf{a} \times \mathbf{a}$$

By definition, the cross product of a vector by itself is 0, since they are colinear.

Problem 1.3.2

PROMPT

Prove de identity

$$\epsilon_{ijk} \epsilon_{mnk} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}$$

SOLUTION

$$\epsilon_{ijk} \epsilon_{mnk}$$

The product of two Levi-Civita tensors can be written as the determinant of a 3×3 matrix of Kronecker deltas:

$$\epsilon_{ijk} \epsilon_{mnp} = \det \begin{pmatrix} \delta_{im} & \delta_{in} & \delta_{ip} \\ \delta_{jm} & \delta_{jn} & \delta_{jp} \\ \delta_{km} & \delta_{kn} & \delta_{kp} \end{pmatrix}$$

We aim to solve the specific case where $p = k$:

$$\epsilon_{ijk} \epsilon_{mnk} = \det \begin{pmatrix} \delta_{im} & \delta_{in} & \delta_{ik} \\ \delta_{jm} & \delta_{jn} & \delta_{jk} \\ \delta_{km} & \delta_{kn} & \delta_{kk} \end{pmatrix}$$

Expanding the determinant along the bottom row, and knowing that $\delta_{kk} = 3$ and $\delta_{ik} \delta_{km} = \delta_{im}$, the matrix simplifies directly to the identity:

$$\begin{aligned} \epsilon_{ijk} \epsilon_{mnk} &= \det \begin{pmatrix} \delta_{im} & \delta_{in} \\ \delta_{jm} & \delta_{jn} \end{pmatrix} \\ &= \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm} \end{aligned}$$

Problem 1.3.3

PROMPT

Using the permutation operator, prove the vector triple product identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$$

SOLUTION

Let $\mathbf{r}$ be the result vector.

$$\mathbf{r} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \rightarrow r_i = \epsilon_{ijk} a_j d_k$$

For $d_k = \epsilon_{kmn} b_m c_n$

Therefore

$$r_i = \epsilon_{ijk} a_j \epsilon_{kmn} b_m c_n$$

From the property obtained from the last problem, and the fact that $\epsilon_{kmn} = \epsilon_{mnk}$, we know that:

$$\epsilon_{ijk} \epsilon_{mnk} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}$$

$$\begin{aligned} r_i &= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) a_j b_m c_n \\ &= a_j b_m c_n \delta_{im} \delta_{jn} - a_j b_m c_n \delta_{in} \delta_{jm} \\ &= b_i (a_j c_j) - c_i (a_j b_j) \end{aligned}$$

Translating the scalar components back to vector notation, where $(a_j c_j) = (\mathbf{a} \cdot \mathbf{c})$:

$$\mathbf{r} = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$$

Problem 1.4.1

PROMPT

Given two vectors $\mathbf{a}$ and $\mathbf{b}$, show that

$$(\mathbf{a} \otimes \mathbf{a})(\mathbf{b} \otimes \mathbf{b}) = (\mathbf{a} \cdot \mathbf{b}) \mathbf{a} \otimes \mathbf{b}$$

SOLUTION

Let the tensor $\mathbf{R}$ be the result.

$$\begin{aligned} \mathbf{R} &= (\mathbf{a} \otimes \mathbf{a})(\mathbf{b} \otimes \mathbf{b}) \rightarrow R_{ij} = a_i a_k b_k b_j \\ &= (a_k b_k) a_i b_j \\ R_{ij} &= (a_k b_k) a_i b_j \rightarrow \mathbf{R} = (\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{a} \otimes \mathbf{b} \end{aligned}$$

Problem 1.4.2

PROMPT

Find the abstract expression for

$$\frac{\partial}{\partial \mathbf{v}} (\mathbf{v} \cdot \mathbf{v})$$

SOLUTION

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{v}}(\mathbf{v} \cdot \mathbf{v}) &\rightarrow \frac{\partial}{\partial v_j}(v_i v_i) = 2 \frac{\partial v_i}{\partial v_j} v_i \\
&= 2 \delta_{ij} v_i \\
&= 2 v_j
\end{aligned}$$

Therefore

$$\frac{\partial}{\partial \mathbf{v}}(\mathbf{v} \cdot \mathbf{v}) = 2\mathbf{v}$$

Problem 1.4.3**PROMPT**

Given two second-order tensor $\mathbf{A}$ and $\mathbf{B}$, show that

$$\frac{\partial}{\partial \mathbf{A}}(\mathbf{A} : \mathbf{B}) = \mathbf{B}$$

SOLUTION

In index notation:

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{A}}(\mathbf{A} : \mathbf{B}) &\rightarrow \frac{\partial}{\partial A_{ij}}(A_{mn} B_{mn}) \\
\frac{\partial}{\partial A_{ij}}(A_{mn} B_{mn}) &= \frac{\partial A_{mn}}{\partial A_{ij}} B_{mn} + \frac{\partial B_{mn}}{\partial A_{ij}} A_{mn} \\
&= \delta_{mi} \delta_{nj} B_{mn} \\
&= B_{ij} \\
&\rightarrow \mathbf{B}
\end{aligned}$$

Problem 1.4.4**PROMPT**

Find the expression for

$$\frac{\partial}{\partial \mathbf{a}}(\mathbf{a} \times \mathbf{b})$$

SOLUTION

In index notation,

$$\begin{aligned}\frac{\partial}{\partial \mathbf{a}}(\mathbf{a} \times \mathbf{b}) &\rightarrow \frac{\partial}{\partial a_l}(\epsilon_{ijk} a_j b_k) = \epsilon_{ijk} b_k \delta_{jl} \\ &= \epsilon_{ilk} b_k = -\epsilon_{ikl} b_k\end{aligned}$$

Problem 1.4.5

PROMPT

Compute the derivative, with respect to $\mathbf{A}$, of

$$\mathbf{A} - \frac{1}{3} \text{trace}(\mathbf{A}) \mathbf{I}$$

SOLUTION

$$\begin{aligned}\frac{1}{3} \frac{\partial}{\partial A_{mn}}(A_{ij} - A_{kk} \delta_{ij}) &= \delta_{im} \delta_{jn} - \frac{1}{3} \delta_{ij} \delta_{km} \delta_{kn} \\ &= \delta_{im} \delta_{jn} - \frac{1}{3} \delta_{ij} \delta_{mn} \\ &= \mathbb{I}_{ijmn} - \frac{1}{3} \delta_{ij} \delta_{mn} \\ &\rightarrow \mathbb{I} - \frac{1}{3} \mathbf{I} \otimes \mathbf{I}\end{aligned}$$

Problem 1.5.1

PROMPT

Using index notation, show that

SOLUTION

a) $\mathbf{I} : \mathbf{I} = 3$

Since $\mathbf{A} : \mathbf{B} \rightarrow A_{ij} B_{ij}$, then our solution is:

$$\begin{aligned}\alpha &= \delta_{ij} \delta_{ij} \\ &= \delta_{ii} \\ &= 3\end{aligned}$$

b) $\mathbb{I} : \mathbb{I} = \mathbb{I}$

By definition, $\mathbb{I}_{ijkl} = \delta_{ik} \delta_{jl}$. Thus

$$\begin{aligned}
\mathbb{I}_{ijkl}\mathbb{I}_{klmn} &= (\delta_{ik}\delta_{jl})(\delta_{km}\delta_{ln}) \\
&= \delta_{im}\delta_{jn} \\
&= \mathbb{I}
\end{aligned}$$

c) $\mathbb{I}^{\text{sym}} : \mathbb{I}^{\text{sym}} = \mathbb{I}^{\text{sym}}$

$$\begin{aligned}
\mathbb{I}_{ijmn}^{\text{sym}}\mathbb{I}_{mnkl}^{\text{sym}} &= \frac{1}{2}(\delta_{im}\delta_{jn} + \delta_{in}\delta_{jm}) \cdot \frac{1}{2}(\delta_{mk}\delta_{nl} + \delta_{ml}\delta_{nk}) \\
&= \frac{1}{4}(\delta_{im}\delta_{jn}\delta_{mk}\delta_{nl} + \delta_{im}\delta_{jn}\delta_{ml}\delta_{nk} + \delta_{in}\delta_{jm}\delta_{mk}\delta_{nl} + \delta_{in}\delta_{jm}\delta_{ml}\delta_{nk}) \\
&= \frac{1}{4}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} + \delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl}) \\
&= \frac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \\
&= \mathbb{I}_{ijkl}^{\text{sym}}
\end{aligned}$$

Problem 1.5.2

PROMPT

Considering the second-order tensor $\mathbf{A}$, using index notation, show that

SOLUTION

a) $(\mathbf{I} \otimes \mathbf{I}) : \mathbf{A} = \text{trace}(\mathbf{A})\mathbf{I}$ In index notation, the left side is:

$$(\delta_{ij}\delta_{kl}) \cdot A_{kl} = \delta_{ij} \cdot A_{kk}$$

b) $\mathbf{A}^{\text{anti}} = \frac{1}{2}(\mathbf{A} - \mathbf{A}^T)$

By definition, $\mathbf{A}^{\text{anti}}$ is:

$$\begin{aligned}
\mathbb{I}^{\text{anti}} : \mathbf{A} &\rightarrow \mathbb{I}_{ijkl}^{\text{anti}} A_{kl} \\
&= \frac{1}{2}(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}) A_{kl} \\
&= \frac{1}{2}(\delta_{ik}\delta_{jl} A_{kl} - \delta_{il}\delta_{jk} A_{kl}) \\
&= \frac{1}{2}(A_{ij} - A_{ji})
\end{aligned}$$

Problem 1.6.1

PROMPT

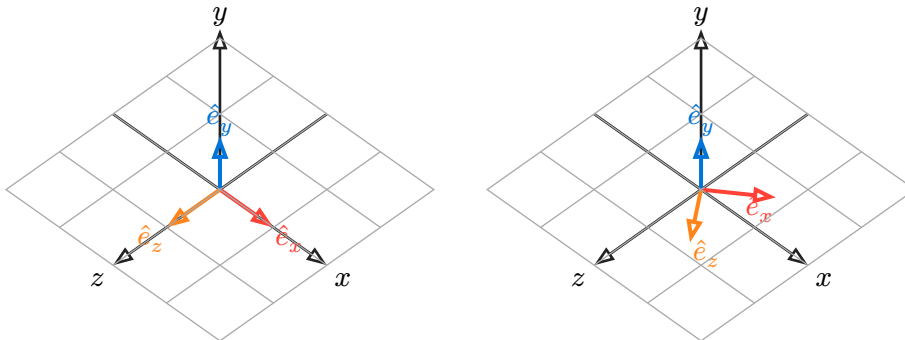
Given the second-order tensor

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 0 & 4 & 0 \\ -1 & 0 & 2 \end{pmatrix}$$

find its components in a new coordinate system obtained by

SOLUTION

a) Rotated by 36.87° around the y-axis.



Thus the new basis vector become

$$\hat{e}_x = \begin{pmatrix} 0.803 \\ 0 \\ -0.596 \end{pmatrix} \hat{e}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \hat{e}_z = \begin{pmatrix} 0.596 \\ 0 \\ 0.803 \end{pmatrix}$$

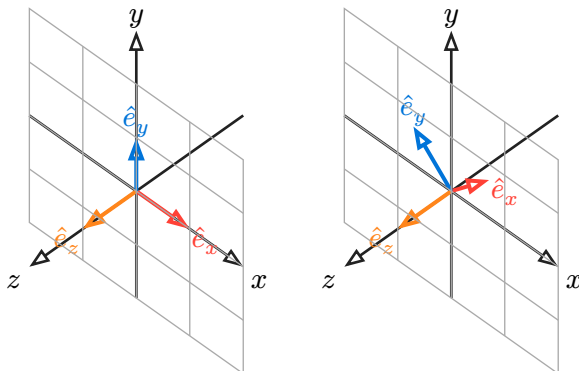
Thus

$$R = \begin{pmatrix} 0.803 & 0 & 0.596 \\ 0 & 1 & 0 \\ -0.596 & 0 & 0.803 \end{pmatrix}$$

By definition, $A' = R^T A R \rightarrow A'_{ij} = R_{ki} A_{kl} R_{lj}$

$$A' = \begin{pmatrix} 3.60189 & 0 & 0.1893 \\ 0 & 4 & 0 \\ 0.1893 & 0 & 1.39811 \end{pmatrix}$$

b) Rotated by 45° around the z-axis.



Thus the new basis vectors become:

$$\hat{e}_x = \begin{pmatrix} 0.707 \\ 0.707 \\ 0 \end{pmatrix} \hat{e}_y = \begin{pmatrix} -0.707 \\ 0.707 \\ 0 \end{pmatrix} \hat{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Thus

$$\mathbf{R} = \begin{pmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

By definition, $\mathbf{A}' = \mathbf{R}^T \mathbf{A} \mathbf{R} \rightarrow A'_{ij} = R_{ki} A_{kl} R_{lj}$

$$\mathbf{A}' = \begin{pmatrix} 3.5 & 0.5 & -0.70711 \\ 0.5 & 3.5 & 0.70711 \\ -0.70711 & 0.70711 & 2 \end{pmatrix}$$

Problem 1.7.1

PROMPT

Given the second-order tensor $\mathbf{A}$

$$\mathbf{A} = \begin{pmatrix} -4.216 & 0 & -2.688 \\ 0 & 12.5 & 0 \\ -2.688 & 0 & 4.216 \end{pmatrix}$$

Evaluate

SOLUTION

a) The eigenvalues and eigenvectors

To obtain the eigenvalues, we must solve

$$\det \left(\begin{pmatrix} -4.216 & 0 & -2.688 \\ 0 & 12.5 & 0 \\ -2.688 & 0 & 4.216 \end{pmatrix} - \mathbf{I}\lambda \right) = 0$$

$$\det \left(\begin{pmatrix} -4.216 - \lambda & 0 & -2.688 \\ 0 & 12.5 - \lambda & 0 \\ -2.688 & 0 & 4.216 - \lambda \end{pmatrix} \right) = 0$$

Thus

$$(\lambda^2 - 4.216^2)(12.5 - \lambda) - (12.5 - \lambda)(2.688^2) = 0$$

$$(12.5 - \lambda)(\lambda^2 - 25) = 0$$

$$(12.5 - \lambda)(\lambda - 5)(\lambda + 5) = 0$$

Thus

$$\lambda = \begin{pmatrix} 12.5 \\ 5 \\ -5 \end{pmatrix}$$

For $\lambda_1 = 12.5$, we must solve

$$\begin{pmatrix} -16.716 & 0 & -2.688 \\ 0 & 0 & 0 \\ -2.688 & 0 & -8.284 \end{pmatrix} \begin{pmatrix} v_{1,1} \\ v_{1,2} \\ v_{1,3} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The only unit vector that solves the equation above is

$$v_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

For $\lambda_2 = 5$, we must solve

$$\begin{pmatrix} -9.216 & 0 & -2.688 \\ 0 & 7.5 & 0 \\ -2.688 & 0 & -0.784 \end{pmatrix} \begin{pmatrix} v_{2,1} \\ v_{2,2} \\ v_{2,3} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

It is clear that $v_{2,2}$ must be null for the solution to be achievable, Thus

$$\begin{pmatrix} -9.216 & -2.688 \\ -2.688 & -0.784 \end{pmatrix} \begin{pmatrix} v_{2,1} \\ v_{2,3} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

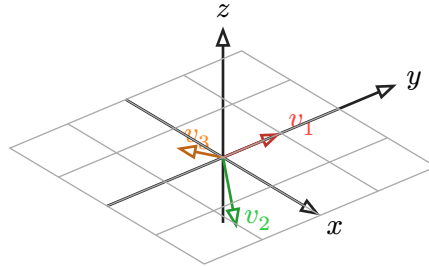
Assuming $v_{2,1} = 1 \rightarrow v_{2,3} = -3.429$

Thus the second eigenvector is

$$v_2 = \begin{pmatrix} 1 \\ 0 \\ -3.429 \end{pmatrix} \rightarrow v_2 = \begin{pmatrix} 0.28 \\ 0 \\ -0.96 \end{pmatrix}$$

To obtain the third eigenvalue, it will be the vector ortogonal to the previous:

$$v_3 = v_1 \times v_2 = \det \left(\begin{pmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ 0 & 1 & 0 \\ 0.28 & 0 & -0.96 \end{pmatrix} \right) = \begin{pmatrix} -0.96 \\ 0 \\ -0.28 \end{pmatrix}$$



b) The principal invariants I_1 , I_2 and I_3

We know that

$$(12.5 - \lambda)(\lambda - 5)(\lambda + 5) = 0$$

Which can be written as

$$\lambda^3 - 12.5\lambda^2 - 25\lambda + 312.5 = 0$$

Which implies

$$\mathbf{I}_A = \begin{pmatrix} 12.5 \\ -25 \\ -312.5 \end{pmatrix}$$

Problem 1.8.1

PROMPT

Calculate $\nabla \cdot \mathbf{u}$ for the following vector fields assuming α and β are constants

SOLUTION

a) $u_r = u_\theta = 0, u_z = \alpha + \beta r^2$

The divergence in cylindrical coordinates is defined by

$$\nabla \cdot \mathbf{f} = \frac{1}{r} \frac{\partial}{\partial r}(r f_r) + \frac{1}{r} \frac{\partial f_\theta}{\partial \theta} + \frac{\partial f_z}{\partial z}$$

Thus, simply:

$$\nabla \cdot \mathbf{u} = \frac{1}{r} \frac{\partial}{\partial r}(r \cdot 0) + \frac{1}{r} \frac{\partial 0}{\partial \theta} + \frac{\partial}{\partial z}(\alpha + \beta r^2) = 0$$

b) $u_r = \frac{\sin(\theta)}{r}, u_z = u_\theta = 0$

$$\begin{aligned} \nabla \cdot \mathbf{u} &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \left(\frac{\sin(\theta)}{r} \right) \right) + \frac{1}{r} \frac{\partial 0}{\partial \theta} + \frac{\partial 0}{\partial z} \\ &= \frac{1}{r} \frac{\partial}{\partial r}(\sin(\theta)) \\ &= 0 \end{aligned}$$

c) $u_r = r^2 \sin(\theta), u_\theta = r^2 \cos(\theta), u_z = 0$

$$\begin{aligned} \nabla \cdot \mathbf{u} &= \frac{1}{r} \frac{\partial}{\partial r}(r(r^2 \sin(\theta))) + \frac{1}{r} \frac{\partial}{\partial \theta}(r^2 \cos(\theta)) + \frac{\partial 0}{\partial z} \\ &= \frac{\sin(\theta)}{r} \frac{\partial r^3}{\partial r} + r \frac{\partial \cos(\theta)}{\partial \theta} \\ &= 3r \sin(\theta) - r \sin(\theta) \\ &= 2r \sin(\theta) \end{aligned}$$

Problem 1.8.2

PROMPT

Calculate $\nabla \cdot \mathbf{u}$ for the vector field given in spherical coordinates assuming α and β as constants.

$$\mathbf{u}_\rho = \alpha\rho + \frac{\beta}{\rho^2}, \mathbf{u}_\theta = \mathbf{u}_\varphi = 0$$

SOLUTION

For spherical coordinates, where f_ρ is the only non null direction, the divergence is defined by

$$\nabla \cdot \mathbf{f} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} (\rho^2 f_\rho)$$

Thus

$$\begin{aligned} \nabla \cdot \mathbf{u} &= \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left[\rho^2 \left(\alpha\rho + \frac{\beta}{\rho^2} \right) \right] \\ &= \frac{1}{\rho^2} \frac{\partial}{\partial \rho} [\alpha\rho^3 + \beta] \\ &= \frac{1}{\rho^2} 3\alpha\rho^2 \\ &= 3\alpha \end{aligned}$$

Problem 1.8.3

PROMPT

Demonstrate that $\frac{\partial \hat{e}_\rho}{\partial \theta} = \sin(\varphi) \hat{e}_\theta$ in spherical coordinates.

SOLUTION

It is known that

$$\hat{e}_\rho = \sin(\varphi) \cos(\theta) \hat{e}_x + \sin(\varphi) \sin(\theta) \hat{e}_y + \cos(\varphi) \hat{e}_z$$

Thus, the derivative is obtained with

$$\begin{aligned} \frac{\partial \hat{e}_\rho}{\partial \theta} &= -\sin(\varphi) \sin(\theta) \hat{e}_x + \sin(\varphi) \cos(\theta) \hat{e}_y \\ &= \sin(\varphi) (-\sin(\theta) \hat{e}_x + \cos(\theta) \hat{e}_y) \\ &= \sin(\varphi) \hat{e}_\theta \end{aligned}$$

Problem 2.1.1

PROMPT

A body is subjected to the displacement field

$$\mathbf{u} = (X_1^2 + X_2^2)\hat{\mathbf{e}}_1 + X_1 X_2 \hat{\mathbf{e}}_2$$

Determine the Green–Lagrange strain tensor $\mathbf{E}$, the infinitesimal strain tensor $\boldsymbol{\varepsilon}$, and the infinitesimal rotation tensor $\boldsymbol{\omega}$.

SOLUTION

In terms of displacement gradient, the Green–Lagrange tensor is

$$\mathbf{E} = \frac{1}{2}(\nabla_{\mathbf{X}}\mathbf{u} + \nabla_{\mathbf{X}}\mathbf{u}^T + \nabla_{\mathbf{X}}\mathbf{u}^T \nabla_{\mathbf{X}}\mathbf{u})$$

The displacement gradient is

$$\nabla_{\mathbf{X}}\mathbf{u} = \begin{pmatrix} 2X_1 & 2X_2 \\ X_2 & X_1 \end{pmatrix}.$$

Its transpose is

$$\nabla_{\mathbf{X}}\mathbf{u}^T = \begin{pmatrix} 2X_1 & X_2 \\ 2X_2 & X_1 \end{pmatrix}.$$

Therefore, the Green–Lagrange strain tensor is

$$\mathbf{E} = \frac{1}{2} \begin{pmatrix} 4X_1 + 4X_1^2 + X_2^2 & 3X_2 + 5X_1 X_2 \\ 3X_2 + 5X_1 X_2 & 2X_1 + X_1^2 + 4X_2^2 \end{pmatrix}.$$

Equivalently,

$$\mathbf{E} = \begin{pmatrix} 2X_1 + 2X_1^2 + \frac{1}{2}X_2^2 & \frac{3}{2}X_2 + \frac{5}{2}X_1 X_2 \\ \frac{3}{2}X_2 + \frac{5}{2}X_1 X_2 & X_1 + \frac{1}{2}X_1^2 + 2X_2^2 \end{pmatrix}.$$

The infinitesimal strain tensor is obtained by neglecting the quadratic term:

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla_{\mathbf{X}}\mathbf{u} + \nabla_{\mathbf{X}}\mathbf{u}^T) = \frac{1}{2} \begin{pmatrix} 4X_1 & 3X_2 \\ 3X_2 & 2X_1 \end{pmatrix}.$$

Hence,

$$\boldsymbol{\varepsilon} = \begin{pmatrix} 2X_1 & \frac{3}{2}X_2 \\ \frac{3}{2}X_2 & X_1 \end{pmatrix}.$$

Finally, the infinitesimal rotation tensor is the antisymmetric part of the displacement gradient:

$$\boldsymbol{\omega} = \frac{1}{2}(\nabla_{\mathbf{X}}\mathbf{u} - \nabla_{\mathbf{X}}\mathbf{u}^T) = \frac{1}{2} \begin{pmatrix} 0 & X_2 \\ -X_2 & 0 \end{pmatrix}.$$

Therefore,

$$\omega = \begin{pmatrix} 0 & \frac{1}{2}X_2 \\ -\frac{1}{2}X_2 & 0 \end{pmatrix}.$$

Problem 2.2.1

PROMPT

Determine whether the following tensor field is a compatible infinitesimal strain field:

$$\varepsilon = \begin{pmatrix} kX_2^2 & kX_1 & 0 \\ kX_1 & kX_2 & 0 \\ 0 & 0 & kX_2^2 \end{pmatrix}$$

SOLUTION

Firstly, assuming $k \neq 0$:

$$\frac{\partial^2 \varepsilon_{11}}{\partial X_2^2} + \frac{\partial^2 \varepsilon_{22}}{\partial X_1^2} - 2 \frac{\partial^2 \varepsilon_{12}}{\partial X_1 \partial X_2} = 2k + 0 + 0 \neq 0$$

Thus, with the first test, the tensor field is incompatible.

Problem 2.2.2

PROMPT

Show that the tensor field below is compatible:

$$\varepsilon = \begin{pmatrix} 2kX_1 & \frac{3}{2}kX_2 & 0 \\ \frac{3}{2}kX_2 & kX_1 & 0 \\ 0 & 0 & kX_3 \end{pmatrix}$$

SOLUTION

It is simple to observe two facts about the tensor ε :

1. The tensor is **symmetric** ($\varepsilon_{ij} = \varepsilon_{ji}$);
2. ε_{ij} is, at **most**, a linear equation.

Thus, all second order derivatives are null, respecting the Saint-Venant compatibility equations.

Problem 2.3.1

PROMPT

Given the displacement field

$$\mathbf{u} = k(X_1^2 + X_2^2)t\hat{e}_1 + kX_1X_2\hat{e}_2 + kX_1X_3\hat{e}_3$$

where $k = 10^{-4}$, determine the volumetric strain at point $\mathbf{X} = (2, 1, 1)$ for $t = 2$.

SOLUTION

To define the relative change in the volume of a material element, we may calculate

$$\varepsilon_v = \frac{\Delta(dV)}{dV} \approx \text{trace}(\varepsilon) = \varepsilon_{ii} = \frac{\partial u_i}{\partial X_i} = \nabla \cdot \mathbf{u}$$

Thus, simply:

$$\begin{aligned}\varepsilon_v &= \frac{\partial(kt(X_1^2 + X_2^2))}{\partial X_1} + \frac{\partial(kX_1X_2)}{\partial X_2} + \frac{\partial(kX_1X_3)}{\partial X_3} = \\ &= 2k(t = 2)X_1 + kX_1 + kX_1 \\ &= 6kX_1 = 12 \cdot 10^{-4}\end{aligned}$$

Problem 2.4.1

PROMPT

Given the tensor field

$$\varepsilon = \begin{pmatrix} 4kX_1 & 3kX_2 & 0 \\ 3kX_2 & 2kX_1 & 0 \\ 0 & 0 & 2kX_3 \end{pmatrix}$$

where $k = 10^{-4}$, determine the principal strains and directions at $\mathbf{X} = (1, 1, 1)$. Compare the two in-plane eigenvalues with the values obtained from the 2D formula.

SOLUTION

At $\mathbf{X} = (1, 1, 1)$, the tensor field becomes

$$\varepsilon = \begin{pmatrix} 4 & 3 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \cdot k$$

Thus, to obtain the eigenvalues ($\lambda' = \frac{\lambda}{k}$):

$$\begin{aligned}\lambda &= \det \begin{pmatrix} 4 - \lambda' & 3 & 0 \\ 3 & 2 - \lambda' & 0 \\ 0 & 0 & 2 - \lambda' \end{pmatrix} \\ &= (4 - \lambda')(2 - \lambda')^2 - 9(2 - \lambda') = 0\end{aligned}$$

Which equates to

$$\lambda'^3 - 8\lambda'^2 + 11\lambda' + 2 = 0$$

The solutions are

$$\lambda = \begin{pmatrix} 2 \\ 3 - \sqrt{10} \\ 3 + \sqrt{10} \end{pmatrix} k$$

The first eigenvector can be obtained with

$$\begin{pmatrix} 2 & 3 & 0 \\ 3 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} v_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Which the only unitary solution is

$$v_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

For the second eigenvector:

$$\begin{pmatrix} 1 + \sqrt{10} & 3 & 0 \\ 3 & -1 + \sqrt{10} & 0 \\ 0 & 0 & -1 + \sqrt{10} \end{pmatrix} v_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 + \sqrt{10} & 3 \\ 3 & -1 + \sqrt{10} \end{pmatrix} v_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Assuming $v_{2,1} = 1$:

$$\begin{aligned}3 + (-1 + \sqrt{10})v_{2,2} &= 0 \\ v_{2,2} &= -1.387\end{aligned}$$

Then, the unitary solution is:

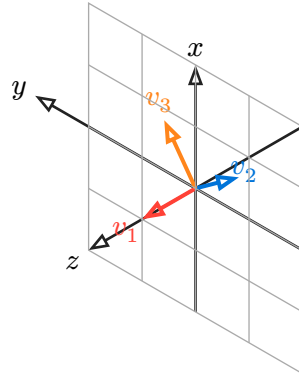
$$v_2 = \begin{pmatrix} 0.585 \\ -0.811 \\ 0 \end{pmatrix}$$

The third, by obtaining the cross product, is:

$$\mathbf{v}_3 = \begin{pmatrix} 0.811 \\ 0.585 \\ 0 \end{pmatrix}$$

Thus, the eigen vectors can be defined by the tensor $\mathbf{Q}$:

$$\mathbf{Q} = \begin{pmatrix} | & | & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 0 & 0.585 & 0.811 \\ 0 & -0.811 & 0.585 \\ 1 & 0 & 0 \end{pmatrix}$$



,

Using the 2D equation:

$$\begin{aligned} \varepsilon_{1;2} &= \frac{1}{2}(4k + 2k) \pm \sqrt{\left(\frac{4k - 2k}{2}\right)^2 + (3k)^2} \\ &= 3k \pm \sqrt{k^2 + 9k^2} \\ &= (3 \pm \sqrt{10})k \end{aligned}$$

Which reproduces the two in-plane eigenvalues, with an out-of-plane strain of $2k$.

Note: The eigenvalues should have been placed in the order $\lambda_1 > \lambda_2 > \lambda_3$ but I'm too lazy to change it now it's like 23h15.

Problem 2.4.2

PROMPT

Given the displacement field

$$\mathbf{u} = k(X_1^2 + X_2^2)t\hat{\mathbf{e}}_1 + kX_1X_2\hat{\mathbf{e}}_2 + kX_1X_3\hat{\mathbf{e}}_3$$

where $k = 10^{-4}$, determine the principal strains and directions at $\mathbf{X} = (2, 1, 1)$ for $t = 1$ and $t = 2$.

SOLUTION

The tensor field is defined by

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} \right)$$

Thus,

$$\varepsilon(X_1, X_2, X_3, t) = \begin{pmatrix} 2X_1t & X_2 + 2X_2t & X_3 \\ X_2 + 2X_2t & X_1 & 0 \\ X_3 & 0 & X_1 \end{pmatrix} \cdot k$$

Which, for $(2, 1, 1)$:

$$\varepsilon(t) = \begin{pmatrix} 4t & 1 + 2t & 1 \\ 1 + 2t & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

For $t = 1$

$$\varepsilon = \begin{pmatrix} 4 & 3 & 1 \\ 3 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

The characteristic polynomial for this tensor field is then

$$\lambda^3 - 8\lambda^2 + 10\lambda + 4 = 0$$

Thus the principal strains are:

$$\lambda = \begin{pmatrix} 3 + \sqrt{11} \\ 2 \\ 3 - \sqrt{11} \end{pmatrix} k$$

With eigenvectors:

$$Q = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 4.31662 & 0.0 & -2.31662 \\ 3.0 & -1.0 & 3.0 \\ 1.0 & 3.0 & 1.0 \end{pmatrix} = \begin{pmatrix} 0.8067 & 0 & -0.5910 \\ 0.5606 & -0.3162 & 0.7653 \\ 0.1868 & 0.9487 & 0.2551 \end{pmatrix}$$

Similarly, for $t = 2$:

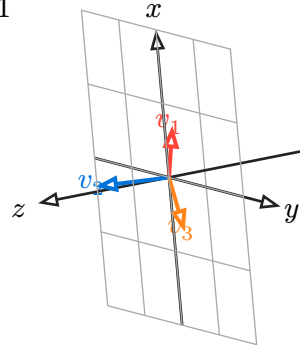
$$\varepsilon = \begin{pmatrix} 8 & 5 & 1 \\ 5 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

The principal strains are:

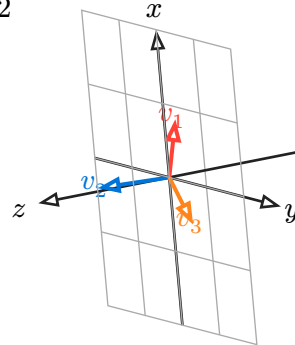
$$\lambda = \begin{pmatrix} 5 + \sqrt{35} \\ 2 \\ 5 - \sqrt{35} \end{pmatrix} k$$

$$Q = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 8.91608 & 0.0 & -2.91608 \\ 5.0 & -1.0 & 5.0 \\ 1.0 & 5.0 & 1.0 \end{pmatrix} = \begin{pmatrix} 0.8685 & 0.0 & -0.4964 \\ 0.4870 & -0.1961 & 0.8511 \\ 0.0974 & 0.9806 & 0.1702 \end{pmatrix}$$

$t = 1$



$t = 2$



Problem 2.4.3

PROMPT

Given the infinitesimal strain tensor ε , find the principal strains and directions. Consider $k = 10^{-4}$.

$$\varepsilon = k \begin{pmatrix} 1 & 0.5 & 0 \\ 0.5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

SOLUTION

To obtain the eigenvalues, the following expression must be solved:

$$\det \begin{pmatrix} 1 - \lambda' & 0.5 & 0 \\ 0.5 & 1 - \lambda' & 0 \\ 0 & 0 & 1 - \lambda' \end{pmatrix} = 0$$

$$(1 - \lambda')^3 - 0.5^2(1 - \lambda') = 0$$

$$\lambda'^3 - 3\lambda'^2 + 2.75\lambda' - 0.75 = 0$$

For $\lambda' = \frac{\lambda}{k}$.

Which solutions are

$$\lambda = \begin{pmatrix} 3/2 \\ 1 \\ 1/2 \end{pmatrix} k$$

Then:

$$\mathbf{Q} = \begin{pmatrix} | & | & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 0.707 & 0 & -0.707 \\ 0.707 & 0 & 0.707 \\ 0 & 1 & 0 \end{pmatrix}$$

Problem 2.5.1

PROMPT

The cylindrical components ε' were calculated at $(r, \theta, z) = (3, \frac{\pi}{6}, 0)$. Convert them to Cartesian components.

$$\varepsilon' = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & 3 \end{pmatrix} \cdot 10^{-3}$$

SOLUTION

To change from cylindrical to cartesian components, we must invert the following equation

$$\varepsilon' = \mathbf{R}^T \varepsilon \mathbf{R}$$

Multiplying on the left by $\mathbf{R}$, and on the right, by $\mathbf{R}^T$:

$$\mathbf{R} \varepsilon' \mathbf{R}^T = \mathbf{R} \mathbf{R}^T \varepsilon \mathbf{R} \mathbf{R}^T$$

Since $\mathbf{R}$ is orthogonal, $\mathbf{R} \mathbf{R}^T = 1$:

$$\varepsilon = \mathbf{R} \varepsilon' \mathbf{R}^T$$

The transformation tensor for cylindrical coordinates becomes:

$$\mathbf{R} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

At $(r, \theta, z) = (3, \frac{\pi}{6}, 0)$:

$$\mathbf{R} = \begin{pmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Finally

$$\begin{aligned} \varepsilon &= \mathbf{R} \varepsilon' \mathbf{R}^T = \begin{pmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0.866 & 0.5 & 0 \\ -0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot 10^{-3} \\ \varepsilon &= \begin{pmatrix} 2.61603 & -0.0669873 & 1.73205 \\ -0.0669873 & 0.383975 & 1.0 \\ 1.73205 & 1.0 & 3.0 \end{pmatrix} \cdot 10^{-3} \end{aligned}$$